

# AJ C PIPATTANAKUN

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## PERSONAL SUMMARY

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Computer Science graduate with a Minor in Data Science, currently pursuing a master's degree in Computer Science. Experience working on academic and internship projects involving coding, debugging, backend services, API testing, software integration, and technical documentation. Comfortable using multiple high-level programming languages including C++, Java, Python, SQL, and basic C, with working knowledge of object-oriented programming and Object-Oriented Analysis and Design (OOAD).

Hands-on experience includes setting up local development environments, troubleshooting build and runtime issues, testing API health endpoints, validating JSON outputs, and checking database connectivity. Familiar with Windows and basic UNIX/Linux command-line environments, Git/GitHub version control, software testing concepts, regression testing practice, and requirement-based validation.

## CORE SKILLS

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**Programming Languages:** C++, Java, Python, SQL, basic C

**Software Engineering:** Coding, debugging, object-oriented programming, Object-Oriented Analysis and Design (OOAD), software design, and technical documentation

**Software Testing:** Unit testing concepts, regression testing, requirement testing, integration testing, API health checks, and validation of expected system behavior

**Systems & Operating Systems:** Windows 10/11, basic UNIX/Linux command line, environment setup, process checks, and software/hardware integration concepts

**Version Control & Tools:** Git, GitHub, Visual Studio Code, Docker, FastAPI, Node.js, Jupyter Notebooks; familiar with version control workflows similar to Bitbucket

**Communication:** Reading technical documentation, writing clear issue notes, explaining progress, and participating in team updates

## EDUCATION

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**University of Massachusetts Lowell**

*Master of Science in Computer Science*

Expected Spring 2028

*Lowell, MA*

- Relevant Courses: Advanced Computer Science, Software Development, Artificial Intelligence, Data Systems

**University of Massachusetts Lowell**

*Bachelor of Science in Computer Science, Minor in Data Science*

September 2020 - August 2024

*Lowell, MA*

- Relevant Courses: Machine Learning, Data Visualization, Statistical Modeling

## WORK EXPERIENCE

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**Software Engineering Intern**

*Tihranix, Foxborough, MA*

May 2026 – Present

- Set up and validated local development environments using Git, GitHub, Docker, VS Code, Node.js, Python, FastAPI, and TypeScript.
- Troubleshoot setup, build, runtime, and API behavior issues during local application testing.
- Tested backend services by checking API health endpoints, validating JSON responses, and reviewing expected output behavior.
- Practiced software integration by confirming that frontend preview logic, backend services, and PLDM-driven context returned consistent results.
- Used Git and GitHub pull requests for version control, branch management, code review, and team workflow practices.

- Built and maintained a responsive website using HTML, CSS, and JavaScript.
- Updated content, tested page behavior across devices, and maintained basic contact/menu functionality.

## PROJECTS

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**Real-Time Multi-Sensor Monitoring and Fault-Tolerant System** January 2026  
*C++, Python, Linux, TCP/IP, CMake, Google Test*

- Designing a multithreaded C++ application to process simulated real-time temperature, voltage, position, and motion sensor data over TCP/IP.
- Implementing object-oriented sensor interfaces, thread-safe message queues, fault detection, structured logging, and deterministic system-state transitions.
- Developing a Python-based sensor simulator and fault-injection tool to test timeouts, malformed messages, abnormal readings, and communication failures.
- Creating unit, integration, regression, and requirement-based tests using Google Test to verify expected system behavior.
- Configuring a Linux build workflow using CMake, GitHub Actions, and basic runtime diagnostic tools.

**Reliable Distributed Command Processing System** April 2026  
*Java, Python, Linux, TCP/IP, JUnit, Gradle, Docker*

- Designing a Java-based distributed system that exchanges structured commands between a control station and a simulated remote unit over TCP/IP.
- Applying object-oriented analysis and design principles through Command, State, Strategy, and Factory design patterns.
- Implementing message validation, acknowledgements, timeouts, retries, duplicate-message detection, sequence-number checks, and audit logging.
- Developing state-dependent command handling to reject invalid operations and maintain predictable system behavior.
- Creating JUnit unit, integration, regression, and requirement-based tests for message parsing, command execution, invalid state transitions, and network failures.

## CERTIFICATIONS

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**Google IT Support Professional Certificate** Completed: January 2021

- Developed by Google and delivered through Coursera; consists of five courses focused on hands-on IT support training.
- Gained foundational skills in troubleshooting, customer service, networking, operating systems, system administration, and IT security.

**UMass Global Data Science Bootcamp** Completed: August 2024

- Comprehensive data science training with a focus on business insights, structured thinking, predictive modeling, and advanced data visualization.
- Gained expertise in using issue trees, value driver trees, A / B testing, and advanced SQL for business analytics.